

AWS Amplify Deployment Guide for React/Node Project

Objective: To create production build of project and deploy it successfully on AWS Amplify.

Requirements:

1. Node.js Installed
2. AWS Account
3. React/Vite Project
4. Internet Connection

Step 1: Open Project in VS Code

Open your project folder in Visual Studio Code and open terminal inside the project directory.

Step 2: Install Dependencies

```
npm install
```

This command installs all required packages from package.json file.

Step 3: Create Production Build

```
npm run build
```

For React CRA project, this command creates a build folder. For Vite project, this command creates a dist folder.

Step 4: Verify Build Folder

```
build/  
├──  
├── index.html  
├── static/  
└── assets/
```

Check whether build folder contains index.html and static assets.

Step 5: Compress Build Folder

Compress only the contents inside build or dist folder into ZIP file. Do not zip the outer build folder itself.

Correct ZIP Structure:

```
index.html  
static/  
assets/
```

Wrong ZIP Structure:

```
build/  
  index.html
```

Step 6: Login to AWS Console

Open AWS Console in browser and login using AWS credentials.

Step 7: Open AWS Amplify

Search for Amplify service in AWS search bar and open AWS Amplify Dashboard.

Step 8: Deploy Application

Click Deploy an App.

Select Deploy without Git Provider.

Upload build.zip or dist.zip file.

Enter application name.

Click Save and Deploy.

Step 9: Access Live Website

After deployment completes successfully, AWS Amplify generates a public URL to access the hosted application.

React Router 403 Error Solution

Create file:

public/_redirects

Add:

```
/* /index.html 200
```

This configuration solves page refresh and routing issues in React applications.

Important Commands Summary

```
npm install
```

```
npm run build
```

Explanation:

AWS Amplify is a Platform as a Service (PaaS) used to deploy and host web applications. It automatically handles deployment, hosting, scaling and SSL configuration. The build folder generated by React/Vite contains optimized production files used for hosting.

Result: Production build was created successfully and web application was deployed on AWS Amplify.